

# Sultan Alaqli

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Website | LinkedIn | GitHub | Hugging Face | Kaggle | X

## PROFILE

AI/ML Engineer specializing in computer vision, NLP, and AI agents. Build end-to-end, production-ready systems, from multi-model AI pipelines to voice-driven agents and RAG assistants, turning the mathematical foundations of machine learning into reliable applications. Led a 3-day advanced ML training for 45+ participants.

## EDUCATION

### Umm Al-Qura University

Bachelor of Artificial Intelligence

- GPA: 3.97 / 4.00

Sep. 2021 – Jun. 2026

Makkah, Saudi Arabia

## EXPERIENCE

### KAUST Academy

Artificial Intelligence Training Program | Note: Accepted for University of Oxford Summer 2026

Jan. 2026 – Ongoing  
Makkah, Saudi Arabia - Oxford, United Kingdom

- Completed an **80-hour** intensive in Deep Learning and Computer Vision: CNNs, transfer learning, image segmentation, object detection, and data augmentation through hands-on labs.
- Stack:** PyTorch, OpenCV, CNNs, Transfer Learning, Image Processing | [Certificates](#)

### The Education University of Hong Kong

Exchange Student (Summer Program)

Jul. 2025 – Aug. 2025

Hong Kong, China

- Intensive DL/CV / Object Detection:** End-to-end detection workflows via labs covering CNNs, transfer learning, data annotation, YOLO training/inference, and evaluation (mAP, CER).
- SmartGrader:** Built a system that grades scanned handwritten exams end-to-end: YOLOv8 to locate sections, EasyOCR + TrOCR to read handwriting, and an LLM to score by rubric, with a review/edit/export page.
- Stack:** PyTorch, YOLOv8, OCR, LLM prompting, experiment tracking | [Certificate](#)

### Deep Scan Technologies | Company

AI Intern

Mar. 2025 – Jun. 2025

Riyadh, Saudi Arabia

- Hotel Voice Assistant (AI Agent):** Built a voice-driven reservation system with n8n + ElevenLabs that books, cancels, and reschedules stays, orchestrating intent and updating booking data in real time.
- Speech Data:** Assembled and cleaned a large Arabic speech corpus from public sources to train Nabrah (Saudi dialect voice model).
- Stack:** PyTorch, n8n, ElevenLabs, Whisper, GPT, Pandas, Git/GitHub, Kaggle

## PROJECTS

### NeoBalady+: Multi-Model AI for Smart-City Operations | PyTorch, ConvLSTM, VLM, Whisper, RAG, OSMnx

Sep. 2025 – Jun. 2026

- Built a multi-model AI backend (graduation project) for Makkah Municipality's Balady app: a quantized Vision-Language Model classifies photo complaints, writes Arabic descriptions, flags day/night duplicates, sets objective priority over exaggerated reports, and verifies repairs from before/after images; a Saudi-dialect Whisper transcribes voice notes.
- Added a ConvLSTM forecaster on Makkah's real street network to flag infrastructure hotspots before they're reported (with Z-Score anomaly alerts), a Random Forest for dynamic SLA estimation, and a RAG Arabic citizen assistant; ran the full pipeline on a single GPU with interactive Gradio dashboards.

### Saudi Dialect End-of-Utterance (EOU) Detection | BERT, NLP, SaudiBERT | Model

Jan. 2026

- Built a Saudi-dialect end-of-utterance detector for voice agents; fine-tuned **SaudiBERT** on a hybrid dataset (public data + 2,136 ALLaM-generated synthetic edge cases), reaching **77.47%** F1 on the held-out test set.
- [Live Demo](#) | [Dataset](#)

### IronID: On-Device Gym Equipment Recognition | TFLite, MobileNetV3, Knowledge Distillation | GitHub

Oct. 2025 – Jan. 2026

- Built a hybrid Android app that recognizes gym equipment fully on-device, removing cloud latency and privacy risk; benchmarked MobileNetV3-Small, EfficientNet-B0 distillation, and MobileNetV3-Large, selecting the latter for **97%** baseline accuracy at mobile-friendly speed, gated by a 60% confidence threshold.
- Compressed the model via magnitude-based pruning (TF Model Optimization Toolkit) and Float16 quantization, cutting size from 11.37MB to **5.72MB** while reaching **99.02%** accuracy at 30–35ms latency; paired with a GPT-3.5-turbo coach for personalized routines, shipped via GitHub Actions CI/CD.

### Arabic Text Correction (NLP) | PyTorch, AraT5, POS Tagging | GitHub

Feb. 2025 – Mar. 2025

- Fine-tuned AraT5 with POS tagging on QALB-2014 for Arabic grammatical error detection and correction, achieving **81.04** F0.5.

## SKILLS

**Technical Skills :** Machine Learning, Deep Learning, Computer Vision, NLP, LLMs & RAG, AI Agents

**Frameworks & Libraries :** PyTorch, Hugging Face Transformers, Scikit-Learn, OpenCV, NumPy, Pandas, Matplotlib

**Programming & Tools :** Python, Java, Git/GitHub, Gradio, n8n, Google Colab

**Soft Skills :** Problem-Solving, Presentation, Teamwork, Leadership, Adaptability, Creativity

**Languages :** Arabic (Native), English (Fluent)